Double Integrals - Solved Quiz Revision

Areas, Volumes, and Change of Order

Area and volume

For a plane region R,

$$\boxed{\text{Area}(R) = \iint_R 1\,dA.}$$

If $f(x,y) \geq 0$, the volume under

$$z = f(x,y)$$

and above R is

$$\boxed{V = \iint_R f(x,y)\,dA.}$$

If the solid lies between

$$z = u(x,y) \quad \text{and} \quad z = \ell(x,y), \qquad u \geq \ell,$$

then

$$\boxed{V = \iint_R \big(u(x,y) - \ell(x,y)\big)\,dA.}$$

Reading bounds

For vertical slices,

$$R = \{(x,y) : a \leq x \leq b,\ g_1(x) \leq y \leq g_2(x)\}$$

gives

$$\boxed{\iint_R f\,dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\,dx.}$$

For horizontal slices,

$$R = \{(x,y) : c \leq y \leq d,\ h_1(y) \leq x \leq h_2(y)\}$$

gives

$$\boxed{\iint_R f\,dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y)\,dx\,dy.}$$

Change-of-order routine

original bounds

↓

sketch the region

↓

choose the new slice direction

↓

find the new outer-variable range

↓

find the new entry and exit boundaries

If the entry or exit boundary changes partway through the new outer-variable range, **split the integral**.

1. Rectangular regions and iterated integrals

Example 1

Evaluate

$$I = \int_0^2 \int_1^3 (x + 2y)\,dy\,dx.$$

Then check the answer by reversing the order.

Detailed solution

Integrate with respect to y first:

$$\begin{aligned} I &= \int_0^2 \left[xy + y^2\right]_1^3 dx \\ &= \int_0^2 \left((3x+9) - (x+1)\right) dx \\ &= \int_0^2 (2x+8)\, dx \\ &= \left[x^2 + 8x\right]_0^2 \\ &= \boxed{20}. \end{aligned}$$

The region is the rectangle

$$0 \le x \le 2, \qquad 1 \le y \le 3.$$

So the reversed order is

$$I = \int_1^3 \int_0^2 (x + 2y)\, dx\, dy.$$

Now

$$\int_0^2 (x+2y)\, dx = \left[\frac{x^2}{2} + 2yx\right]_0^2 = 2 + 4y.$$

Hence

$$\int_1^3 (2+4y)\, dy = \left[2y + 2y^2\right]_1^3 = 20.$$

Both orders give the same answer.

Common mistake

Reversing order on a rectangle is easy because all four limits remain constants. This is not true for general regions.

2. Volumes

Example 2 - volume under a surface over a rectangle

Find the volume under

$$z = 5 - x - y$$

above the rectangle

$$0 \le x \le 1, \qquad 0 \le y \le 2.$$

Detailed solution

Since

$$5 - x - y \ge 2$$

on the rectangle, the surface is above the xy-plane throughout.
Therefore

$$V = \int_0^1 \int_0^2 (5 - x - y)\, dy\, dx.$$

Integrate with respect to y:

$$\begin{aligned} \int_0^2 (5-x-y)\, dy &= \left[(5-x)y - \frac{y^2}{2}\right]_0^2 \\ &= 10 - 2x - 2 \\ &= 8 - 2x. \end{aligned}$$

Hence

$$V = \int_0^1 (8 - 2x)\, dx = \left[8x - x^2\right]_0^1 = \boxed{7}.$$

Example 3 - volume between two surfaces

Over the unit square

$$R = [0,1] \times [0,1],$$

find the volume between

$$z_{\text{top}} = 4 + x$$

and

$$z_{\text{bottom}} = x^2 + y^2.$$

Detailed solution

The vertical height of the solid is

$$z_{\text{top}} - z_{\text{bottom}} = 4 + x - x^2 - y^2.$$

Therefore

$$V = \int_0^1 \int_0^1 (4 + x - x^2 - y^2)\, dy\, dx.$$

Integrate with respect to y:

$$\int_0^1 (4 + x - x^2 - y^2)\, dy = 4 + x - x^2 - \frac{1}{3}.$$

Thus

$$\begin{aligned} V &= \int_0^1 \left(4 + x - x^2 - \frac{1}{3}\right) dx \\ &= 4 + \frac{1}{2} - \frac{1}{3} - \frac{1}{3} \\ &= \boxed{\frac{23}{6}}. \end{aligned}$$

3. Areas and volumes over general regions

Example 4 - area using vertical slices

Find the area enclosed by

$$y = x^2 \qquad \text{and} \qquad y = 2 - x.$$

Detailed solution

Find the intersections:

$$x^2 = 2 - x \quad \Longrightarrow \quad x^2 + x - 2 = 0.$$

Hence

$$(x + 2)(x - 1) = 0,$$

so

$$x = -2, \qquad x = 1.$$

On this interval,

$$x^2 \leq 2 - x.$$

Therefore

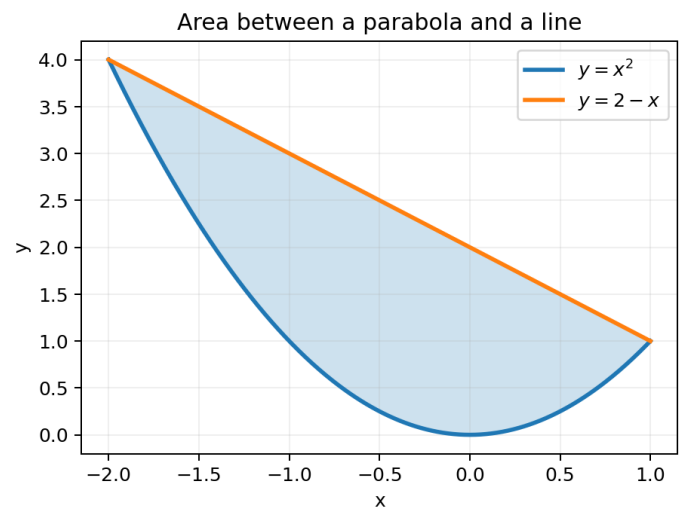
$$A = \int_{-2}^{1} \int_{x^2}^{2-x} 1\, dy\, dx.$$

The inner integral gives

$$2 - x - x^2.$$

Hence

$$\begin{aligned} A &= \int_{-2}^{1} (2 - x - x^2)\, dx \\ &= \left[2x - \frac{x^2}{2} - \frac{x^3}{3}\right]_{-2}^{1} \\ &= \boxed{\frac{9}{2}}. \end{aligned}$$



Example 5 - area using horizontal slices

Find the area bounded by

$$x = y^2 \qquad \text{and} \qquad x = 4.$$

Detailed solution

The curves meet when

$$y^2 = 4,$$

so

$$y = -2, \qquad y = 2.$$

For each fixed y, the horizontal slice runs from

$$x = y^2$$

to

$$x = 4.$$

Therefore

$$A = \int_{-2}^{2} \int_{y^2}^{4} 1 \, dx \, dy.$$

Thus

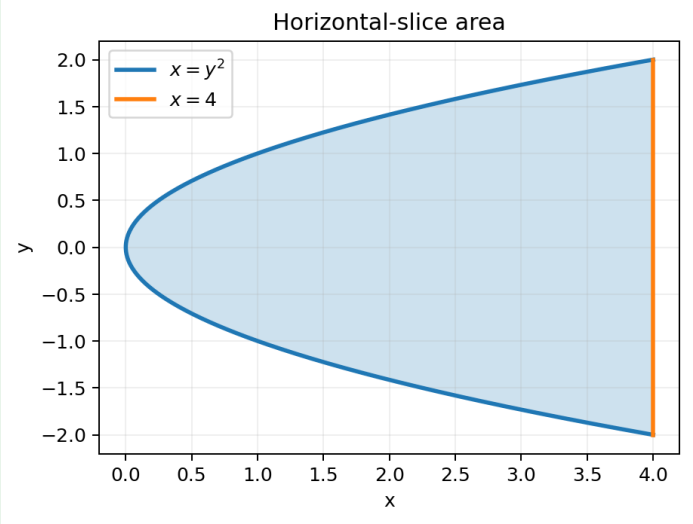
$$A = \int_{-2}^{2} (4 - y^2) \, dy.$$

The integrand is even:

$$A = 2 \int_{0}^{2} (4 - y^2) \, dy.$$

Hence

$$\begin{aligned} A &= 2 \left[4y - \frac{y^3}{3} \right]_0^2 \\ &= 2 \left(8 - \frac{8}{3} \right) \\ &= \boxed{\frac{32}{3}}. \end{aligned}$$



Example 6 - volume over a triangular region

Find the volume under

$$z = 6 - x - y$$

above the triangular region in the first quadrant bounded by

$$x + y = 2.$$

Detailed solution

The base region is

$$0 \le x \le 2, \qquad 0 \le y \le 2 - x.$$

Therefore

$$V = \int_0^2 \int_0^{2-x} (6 - x - y)\, dy\, dx.$$

First integrate in y:

$$\int_0^{2-x} (6 - x - y)\, dy = (6 - x)(2 - x) - \frac{(2 - x)^2}{2}$$

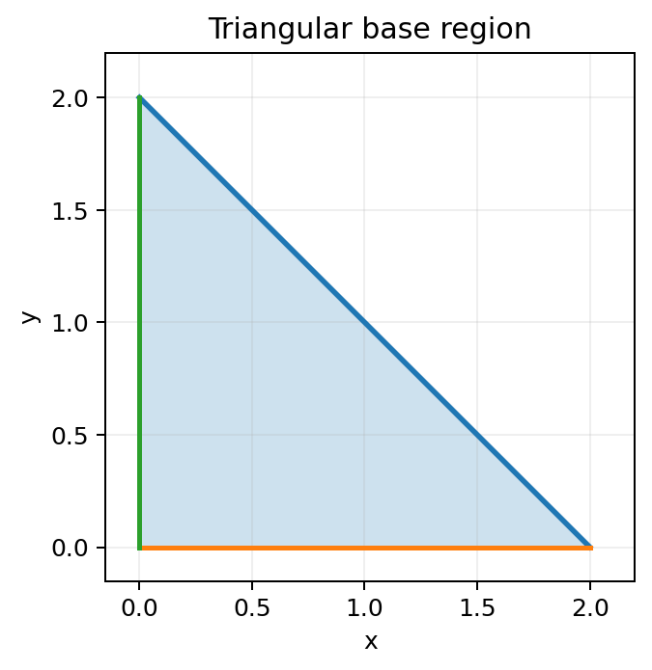
$$= 10 - 6x + \frac{x^2}{2}.$$

Hence

$$V = \int_0^2 \left(10 - 6x + \frac{x^2}{2}\right) dx$$

$$= \left[10x - 3x^2 + \frac{x^3}{6}\right]_0^2$$

$$= \boxed{\frac{28}{3}}.$$



Example 7 - volume between surfaces over a general region

Let

$$R = \{(x, y) : 0 \le x \le 1,\ 0 \le y \le 1 - x\}.$$

Find the volume between

$$z_{\text{top}} = 3 + x \qquad \text{and} \qquad z_{\text{bottom}} = y.$$

Detailed solution

The height is

$$(3 + x) - y.$$

Thus

$$V = \int_0^1 \int_0^{1-x} (3 + x - y)\, dy\, dx.$$

Integrating with respect to y,

$$\int_0^{1-x} (3 + x - y)\, dy = (3 + x)(1 - x) - \frac{(1 - x)^2}{2}.$$

After simplification,

$$(3 + x)(1 - x) - \frac{(1 - x)^2}{2} = \frac{5}{2} - x - \frac{3}{2}x^2.$$

Therefore

$$\begin{aligned} V &= \int_0^1 \left(\frac{5}{2} - x - \frac{3}{2}x^2\right) dx \\ &= \frac{5}{2} - \frac{1}{2} - \frac{1}{2} \\ &= \boxed{\frac{3}{2}}. \end{aligned}$$

4. Change of order - no splitting

Example 8

Change the order of integration in

$$\int_0^3 \int_{x^2}^{3x} f(x,y)\,dy\,dx.$$

Detailed solution

The original region is

$$0 \le x \le 3, \qquad x^2 \le y \le 3x.$$

The curves meet at

$$x^2 = 3x \quad \Longrightarrow \quad x = 0,\ 3.$$

The full y-range is

$$0 \le y \le 9.$$

Solve the boundaries for x:

$$y = 3x \quad \Longrightarrow \quad x = \frac{y}{3},$$

and, in the first quadrant,

$$y = x^2 \quad \Longrightarrow \quad x = \sqrt{y}.$$

For each fixed $y \in [0,9]$, the region runs horizontally from

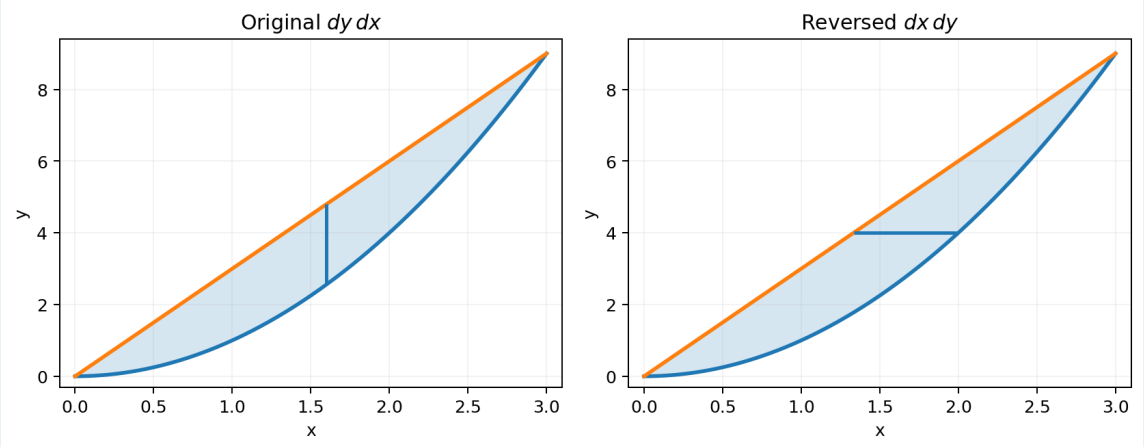
$$x = \frac{y}{3}$$

to

$$x = \sqrt{y}.$$

Therefore

$$\boxed{\int_0^3 \int_{x^2}^{3x} f(x,y)\,dy\,dx = \int_0^9 \int_{y/3}^{\sqrt{y}} f(x,y)\,dx\,dy.}$$



Example 9 - evaluate after reversing

Evaluate

$$I = \int_0^4 \int_{x/2}^{\sqrt{x}} x\,dy\,dx$$

by changing the order first.

Detailed solution

The region is

$$0 \le x \le 4, \qquad \frac{x}{2} \le y \le \sqrt{x}.$$

Rewrite both curves in terms of x:

$$y = \frac{x}{2} \quad \Longrightarrow \quad x = 2y,$$

$$y = \sqrt{x} \quad \Longrightarrow \quad x = y^2.$$

The full y-range is

$$0 \le y \le 2.$$

For fixed y,

$$y^2 \le x \le 2y.$$

Therefore

$$I = \int_0^2 \int_{y^2}^{2y} x \, dx \, dy.$$

Now

$$\int_{y^2}^{2y} x \, dx = \left[\frac{x^2}{2}\right]_{y^2}^{2y} = 2y^2 - \frac{y^4}{2}.$$

Thus

$$\begin{aligned} I &= \int_0^2 \left(2y^2 - \frac{y^4}{2}\right) dy \\ &= \left[\frac{2y^3}{3} - \frac{y^5}{10}\right]_0^2 \\ &= \boxed{\frac{32}{15}}. \end{aligned}$$

5. Change of order - splitting is necessary

Example 10 - fundamental split-region pattern

Reverse the order in

$$\int_0^2 \int_{x/2}^{3-x} f(x,y) \, dy \, dx.$$

Detailed solution

The region satisfies

$$0 \le x \le 2, \qquad \frac{x}{2} \le y \le 3 - x.$$

The boundaries meet at

$$\frac{x}{2} = 3 - x \quad \Longrightarrow \quad x = 2, \qquad y = 1.$$

The region has vertices

$$(0,0), \qquad (0,3), \qquad (2,1).$$

Its full y-range is

$$0 \le y \le 3.$$

For

$$0 \le y \le 1,$$

the right boundary comes from

$$y = \frac{x}{2} \quad \Longrightarrow \quad x = 2y.$$

Thus

$$0 \le x \le 2y.$$

For

$$1 \le y \le 3,$$

the right boundary comes from

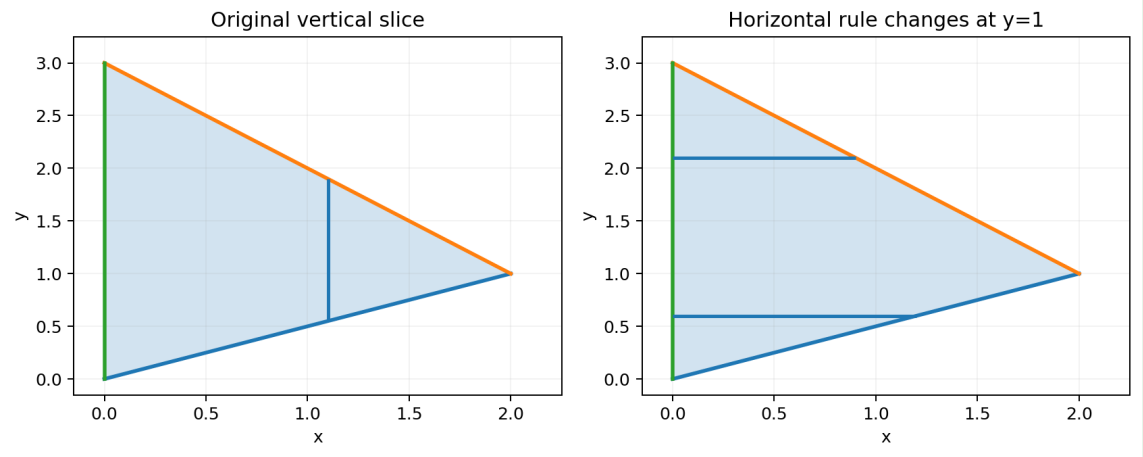
$$y = 3 - x \quad \Longrightarrow \quad x = 3 - y.$$

Thus

$$0 \le x \le 3 - y.$$

Therefore

$$\boxed{\int_0^2 \int_{x/2}^{3-x} f \, dy \, dx = \int_0^1 \int_0^{2y} f \, dx \, dy + \int_1^3 \int_0^{3-y} f \, dx \, dy.}$$



Key idea

The split point is not guessed. It is the y-value where the formula for the new horizontal exit boundary changes.

Example 11 - split at $y = 0$

Evaluate

$$I = \int_0^1 \int_{-x}^{x} (x+1)\,dy\,dx$$

after reversing the order.

Detailed solution

The region is

$$0 \le x \le 1, \qquad -x \le y \le x.$$

For a horizontal slice, the left boundary depends on the sign of y.
If

$$-1 \le y \le 0,$$

then $y \ge -x$ gives

$$x \ge -y.$$

So

$$-y \le x \le 1.$$

If

$$0 \le y \le 1,$$

then $y \le x$ gives

$$x \ge y.$$

So

$$y \le x \le 1.$$

Hence

$$I = \int_{-1}^{0} \int_{-y}^{1} (x+1)\,dx\,dy + \int_0^1 \int_y^1 (x+1)\,dx\,dy.$$

For the first piece,

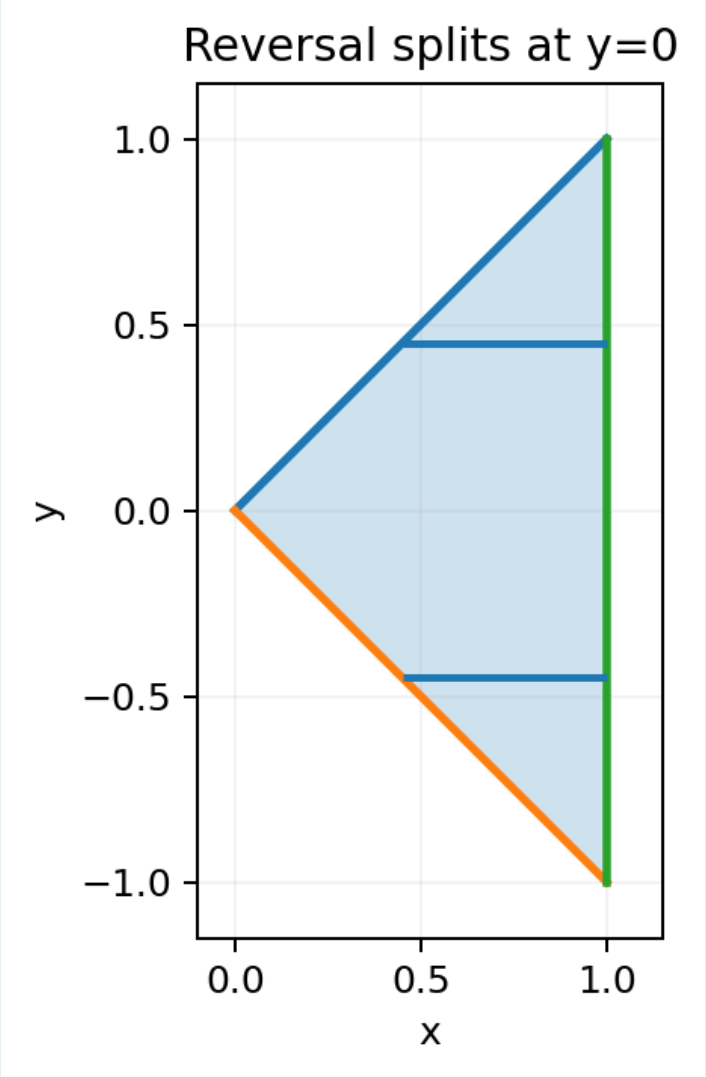
$$\int_{-y}^{1} (x+1)\,dx = \frac{3}{2} + y - \frac{y^2}{2}.$$

For the second,

$$\int_y^1 (x+1)\,dx = \frac{3}{2} - y - \frac{y^2}{2}.$$

Therefore

$$I = \int_{-1}^{0} \left(\frac{3}{2} + y - \frac{y^2}{2}\right) dy + \int_0^1 \left(\frac{3}{2} - y - \frac{y^2}{2}\right) dy$$
$$= \boxed{\frac{5}{3}}.$$



Example 12 - a curved split boundary

Reverse the order in

$$\int_0^1 \int_{x^2}^{2-x} f(x,y)\,dy\,dx.$$

Detailed solution

The region is

$$0 \le x \le 1, \qquad x^2 \le y \le 2-x.$$

The full y-range is

$$0 \le y \le 2.$$

For a horizontal slice, solve the two boundaries:

$$y = x^2 \quad \Longrightarrow \quad x = \sqrt{y},$$

and

$$y = 2 - x \quad \Longrightarrow \quad x = 2 - y.$$

The active right boundary changes when

$$\sqrt{y} = 2 - y.$$

On $0 \le y \le 2$, the relevant solution is

$$y = 1.$$

Thus:
for $0 \le y \le 1$,

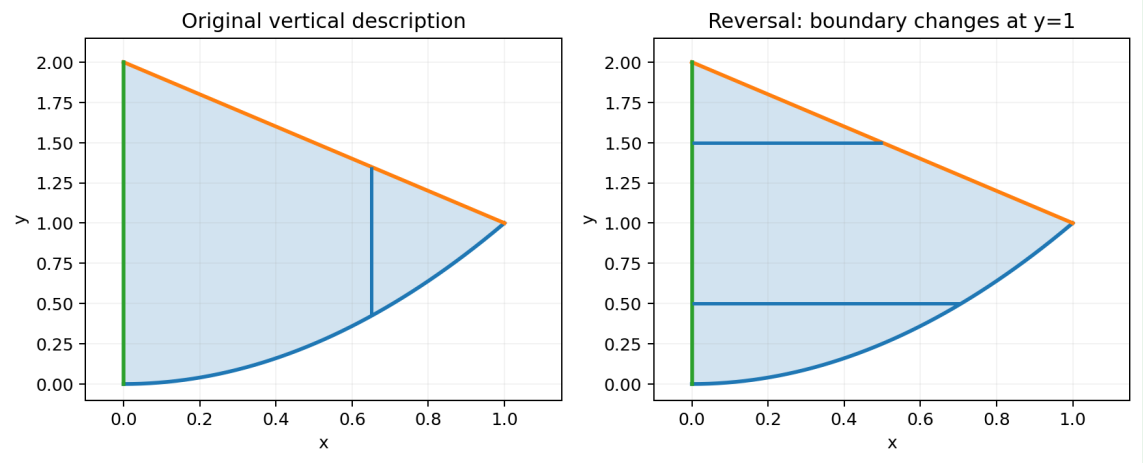
$$0 \le x \le \sqrt{y};$$

for $1 \le y \le 2$,

$$0 \le x \le 2 - y.$$

Therefore

$$\boxed{\int_0^1 \int_{x^2}^{2-x} f \, dy \, dx = \int_0^1 \int_0^{\sqrt{y}} f \, dx \, dy + \int_1^2 \int_0^{2-y} f \, dx \, dy.}$$



6. Reverse the order because the original integral is not elementary

Example 13 - exponential e^{y^3}

Evaluate

$$I = \int_0^1 \int_{\sqrt{x}}^1 e^{y^3}\, dy\, dx.$$

Detailed solution

The original inner integral

$$\int e^{y^3}\, dy$$

has no elementary antiderivative. This is a signal to inspect the region and try the opposite order.
The bounds say

$$0 \le x \le 1, \qquad \sqrt{x} \le y \le 1.$$

The condition

$$\sqrt{x} \le y$$

is equivalent to

$$x \le y^2.$$

The full y-range is

$$0 \le y \le 1.$$

For fixed y,

$$0 \le x \le y^2.$$

Therefore

$$I = \int_0^1 \int_0^{y^2} e^{y^3}\, dx\, dy.$$

Now e^{y^3} is constant with respect to x:

$$\int_0^{y^2} e^{y^3}\, dx = y^2 e^{y^3}.$$

Hence

$$I = \int_0^1 y^2 e^{y^3}\, dy.$$

Let

$$u = y^3, \qquad du = 3y^2\, dy.$$

Then

$$\begin{aligned} I &= \frac{1}{3}\int_0^1 e^u\, du \\ &= \frac{1}{3}[e^u]_0^1 \\ &= \boxed{\frac{e-1}{3}}. \end{aligned}$$

Key idea

The aim of changing order is not merely to rewrite the same integral. Sometimes it changes an impossible-looking inner antiderivative into a simple multiplication by the width of a slice.

Example 14 - trigonometric non-elementary inner integral

Evaluate

$$I = \int_0^1 \int_x^1 \cos(y^2)\, dy\, dx.$$

Detailed solution

The original inner antiderivative

$$\int \cos(y^2)\,dy$$

is not elementary.
The region is

$$0 \le x \le 1, \qquad x \le y \le 1.$$

Equivalently,

$$0 \le y \le 1, \qquad 0 \le x \le y.$$

Thus

$$I = \int_0^1 \int_0^y \cos(y^2)\,dx\,dy.$$

Integrating with respect to x,

$$\int_0^y \cos(y^2)\,dx = y\cos(y^2).$$

Therefore

$$I = \int_0^1 y\cos(y^2)\,dy.$$

Let

$$u = y^2, \qquad du = 2y\,dy.$$

Hence

$$\begin{aligned} I &= \frac{1}{2}\int_0^1 \cos u\,du \\ &= \frac{1}{2}[\sin u]_0^1 \\ &= \boxed{\frac{\sin 1}{2}}. \end{aligned}$$

7. Writing the same region in both orders

Example 15

The region R is bounded by

$$y = x^2, \qquad y = 1, \qquad x = 0$$

in the first quadrant.

(a) Write $\iint_R f\,dA$ in the order $dy\,dx$.

(b) Write it in the order $dx\,dy$.

(c) Use either description to find the area.

Detailed solution

The curve $y = x^2$ meets $y = 1$ at

$$x = 1.$$

(a) Vertical slices
For

$$0 \le x \le 1,$$

the slice begins at

$$y = x^2$$

and ends at

$$y = 1.$$

Therefore

$$\boxed{\iint_R f\,dA = \int_0^1 \int_{x^2}^1 f(x,y)\,dy\,dx.}$$

(b) Horizontal slices
The full y-range is

$$0 \le y \le 1.$$

At fixed y, the region runs from

$$x = 0$$

to

$$x = \sqrt{y}.$$

Hence

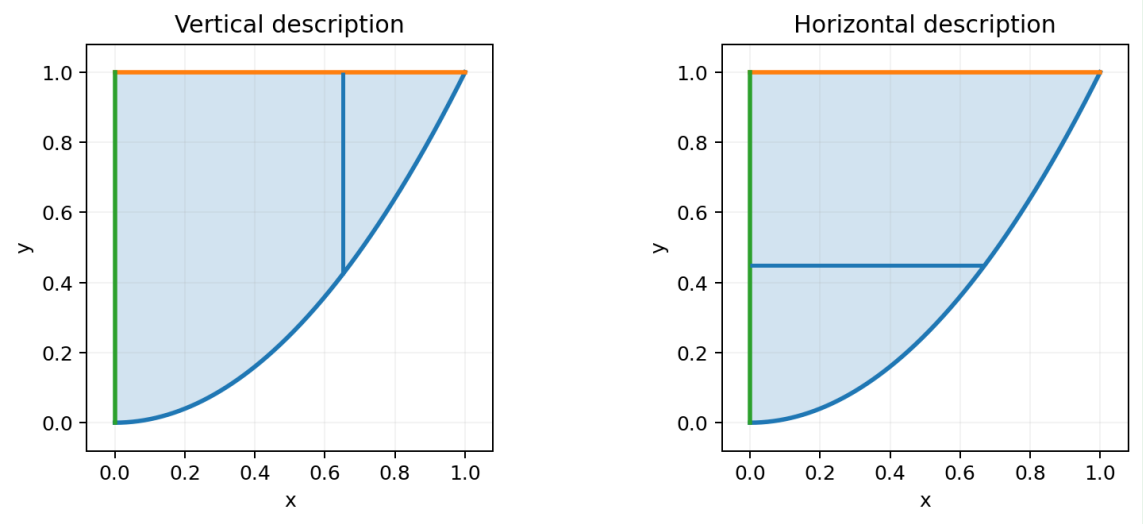
$$\boxed{\iint_R f\,dA = \int_0^1 \int_0^{\sqrt{y}} f(x,y)\,dx\,dy.}$$

(c) Area
Using vertical slices,

$$A = \int_0^1 \int_{x^2}^1 1\,dy\,dx.$$

Thus

$$A = \int_0^1 (1-x^2)\,dx = \left[x - \frac{x^3}{3}\right]_0^1 = \boxed{\frac{2}{3}}.$$



Final checklist

Key idea

Before writing any double integral over a non-rectangular region:

1. sketch the region;
2. find all intersection points;
3. decide whether the inner slices are vertical or horizontal;
4. identify the entry and exit boundary for one typical slice;
5. check whether either boundary changes as the outer variable moves;
6. split the integral if it changes;
7. only then integrate.

Common mistake

Do not "change the order" by merely swapping $dx\,dy$ to $dy\,dx$. The limits belong to the geometry of the region and must be rebuilt from the new slice direction.